

SAFETY DATA SHEET

Sulfur Hexafluoride

SECTION 1 : CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

1.1 Product Details

Product Name : Sulfur Hexafluoride

Chemical Formula : SF₆

1.2 Company Identification

Company's Name : Leeden Gases Sdn. Bhd

Company's Address : Lot PT 5074 & 5075, Jalan Jangur 28/43,
Seksyen 28, Hicom Industrial Estate,
40400 Shah Alam, Selangor.

Emergency telephone Number : 03-55228222 (Hunting Line) / 03-55228288 (Hotline)

SECTION 2 : HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification : Liquefied gas H280

2.2 Label elements

Hazard pictogram :



Signal word : WARNING

Hazard statement : Contains gas under pressure; may explode if heated.
Asphyxiant in high concentrations.

Precautions statements :

- Do not handle until all safety precautions have been read and understood.
- Do not get in eyes, on skin, or on clothing.
- Use and store only outdoors or in a well-ventilated place.
- Use a back flow preventive device in the piping.
- Close valve after each use and when empty.
- Protect from sunlight when ambient temperature exceeds 52°C (125°F)

2.3 Other hazards

Other hazards not contributing to the classification : Asphyxiant in high concentrations.
Contact with liquid may cause cold burns/frostbite

SECTION 3 : COMPOSITION / INFORMATION ON INGREDIENT

Substance/Preparation : Substance.
Components/Impurities : Sulphur hexafluoride.
CAS No : 2551-62-4

SECTION 4 : FIRST AID MEASURES

Inhalation : In high concentrations may cause asphyxiation. Symptoms may include loss of mobility/consciousness. Victim may not be aware of asphyxiation. Remove victim to uncontaminated area wearing self-contained breathing apparatus. Keep victim warm and rested. Call a doctor. Apply artificial respiration if breathing stopped.

Skin/eye contact : For liquid spillage - flush with water for at least 15 minutes In case of frostbite spray with water for at least 15 minutes. Apply a sterile dressing. Obtain medical assistance.

Ingestion : Ingestion is not considered a potential route of exposure.

SECTION 5 : FIRE FIGHTING MEASURES

Specific Hazards : Exposure to fire may cause containers to rupture/explode. Non flammable.

Hazardous Combustion : If involved in a fire the following toxic and/or corrosive Products fumes may be produced by thermal decomposition :
- Hydrogen fluoride.
- Sulphur dioxide.

Suitable Extinguishing Media : All known extinguishants can be used.

Specific Methods : If possible, stop flow of product. Coordinate fire measure to the surrounding fire. Cool endangered containers with water spray jet from a protected position. Do not empty contaminated fire water into drains. Move away from the container and cool with water from a protected position.

Special Protective Equipment : Use self-contained breathing apparatus and chemically for Fire Fighters protective clothing.

SECTION 6 : ACCIDENTAL RELEASE MEASURE

Personal Precautions : Evacuate area. Wear self-contained breathing apparatus when entering area unless atmosphere is proved to be safe. Ensure adequate air ventilation.

Environmental Precautions : Try to stop release. Prevent from entering sewers, basements and workpits, or any place where its accumulation can be dangerous.

Clean-up methods : Ventilate area.

SECTION 7 : HANDLING AND STORAGE

- Handling : Suck back of water into the container must be prevented. Do not allow backfeed into the container. Use only properly specified equipment which is suitable for this product, its supply pressure and temperature. Contact LGSB or your supplier if in doubt. Refer to supplier's handling instructions.
- Storage : Secure cylinders to prevent them falling. Keep container below 50°C in a well ventilated place.

SECTION 8 : EXPOSURE CONTROL AND PERSONAL PROTECTION

8.1 Control Parameter

Chemical Name	ACGIH TLV-TWA (ppm)	OSHA PEL (TWA) (mg/m ³)	NIOSH IDLH (ppm)
Sulfur Hexafluoride 2551-62-4	1,000	6,000	1,000

8.2 Exposure Control

- Appropriate engineering controls : Oxygen detectors should be used when asphyxiating gases may be released. Consider work permit system e.g. for maintenance activities. Systems under pressure should be regularly checked for leakages. Provide adequate general and local exhaust ventilation. Ensure exposure is below occupational exposure limits (where available).
- Hand protection : Wear working gloves when handling gas containers
- Eye protection : Wear safety glasses with side shields or goggles when transfilling or breaking transfer connections. Wear safety glasses with side shields.
- Respiratory protection : Self contained breathing apparatus (SCBA) or positive pressure airline with mask are to be used in oxygen-deficient atmospheres.
- Thermal hazard protection : Wear cold insulating gloves when transfilling or breaking transfer connections. None necessary
- Other information : Wear safety shoes while handling containers. Wear leather safety gloves and safety shoes when handling cylinders.

SECTION 9 : PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Colourless
Physical State	: Gas
Odour	: Odourless
pH Value	: Not applicable
Auto-ignition Temperature	: 0°C
Boiling Point	: -63.9°C
Melting Point	: -50.8°C
Molecular Weight	: 146.05 g/mol
Water Solubility (at 0°C)	: 0.007 vol/vol
Vapor Pressure	: 2100 kPa
Critical Pressure	: 3760 kPa
Relative Density (at 15°C, Air = 1)	: 5.114
Density of Gas (101.3 kPa, 15°C)	: 6.27 kg/m ³
Critical Temperature	: 45.5°C
Flammability Limits	: Non Flammable

SECTION 10 : STABILITY AND REACTIVITY

Reactivity	: No reactivity hazard other than the effects described in sub-sections below.
Chemical Stability	: Stable under normal conditions.
Possibility of Hazardous Reactions	: None.
Conditions to avoid	: None under recommended storage and handling conditions.
Incompatible Materials	: None.
Hazardous Decomposition Products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11 : TOXICOLOGICAL INFORMATION

No known toxicological effects from this product.

SECTION 12 : ECOLOGICAL INFORMATION

No known ecological effects from this product.

SECTION 13 : DISPOSAL INFORMATION

Waste Disposal Method : Do not attempt to dispose of residual or unused quantities. Return cylinder to supplier.

SECTION 14 : TRANSPORT INFORMATION

Shipping Name : Sulfur Hexafluoride
Identification Number : UN 1080
Hazard Class : 2.2

SECTION 15 : REGULATORY INFORMATION

Further national regulations : - Pressure Vessel Regulation.
- Regulations for the prevention of industrial accidents.
SARA Section 311/312 : Immediate (acute) health hazard Sudden release of pressure
Hazard Classes hazard.

SECTION 16 : OTHER INFORMATION

Ensure all national/local regulations are observed. The hazard of asphyxiation is often overlooked and must be stressed during operator training. Before using this product in any new process or experiment, a thorough material compatibility and safety study should be carried out.

Advice

Whilst proper care has been taken in the preparation of this document, no liability for injury or damage resulting from its use can be accepted. Details given in this document are believed to be correct at the time of going to press.